

Lec 22: Network layer numericals + IP header

6/11/23

	Net bits	Host bits	Range
Class A	8	24	0.0.0.0 to 127.255.255.255
Class B	16	16	128.0.0.0 to 191.255.255.255
Class C	24	8	192.0.0.0 to 223.255.255.255

	Net bits	Host bits
Network addr	Specified (As it is)	0...0
Subnet mask	1...1	0...0
Broadcast addr	Specified or 1...1	1...1

Max. # hosts on a network = $2^n - 2$
 n : # of host id bits

Max # networks = 2^k
 k : # of network bits

Problem 3

10.129.0.0/16

... Given

4 labs with # hosts

Lab A

200

8 bits

Lab B

100

7 bits

Lab C

50

6 bits

Lab D

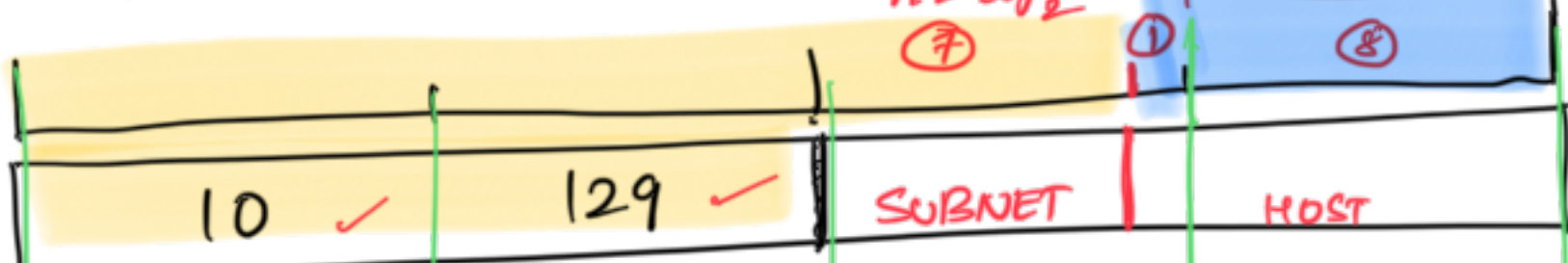
50

6 bits

$$\# \text{ hosts} = 200 + 100 + 50 + 50 = 400$$

$$n = \log_2 400$$

$$n = 9$$



Lab A
200 hosts

10

129

10.129.0.0/24

10.129.0.1 to 10.129.0.254

00000000 0
00000000 0

00000000
11111111

Lab B
100 hosts

10

129

10.129.1.0/25

00000001
00000000

10.129.1.1 to 10.129.1.126

00000000
11111111

Lab C
50 hosts

10

129

10.129.1.128/26

00000001
00000001

10.129.1.129 to 10.129.1.191

10
10
00000000
11111111

Lab D
50 hosts

10

129

10.129.1.192/26

00000001
00000001

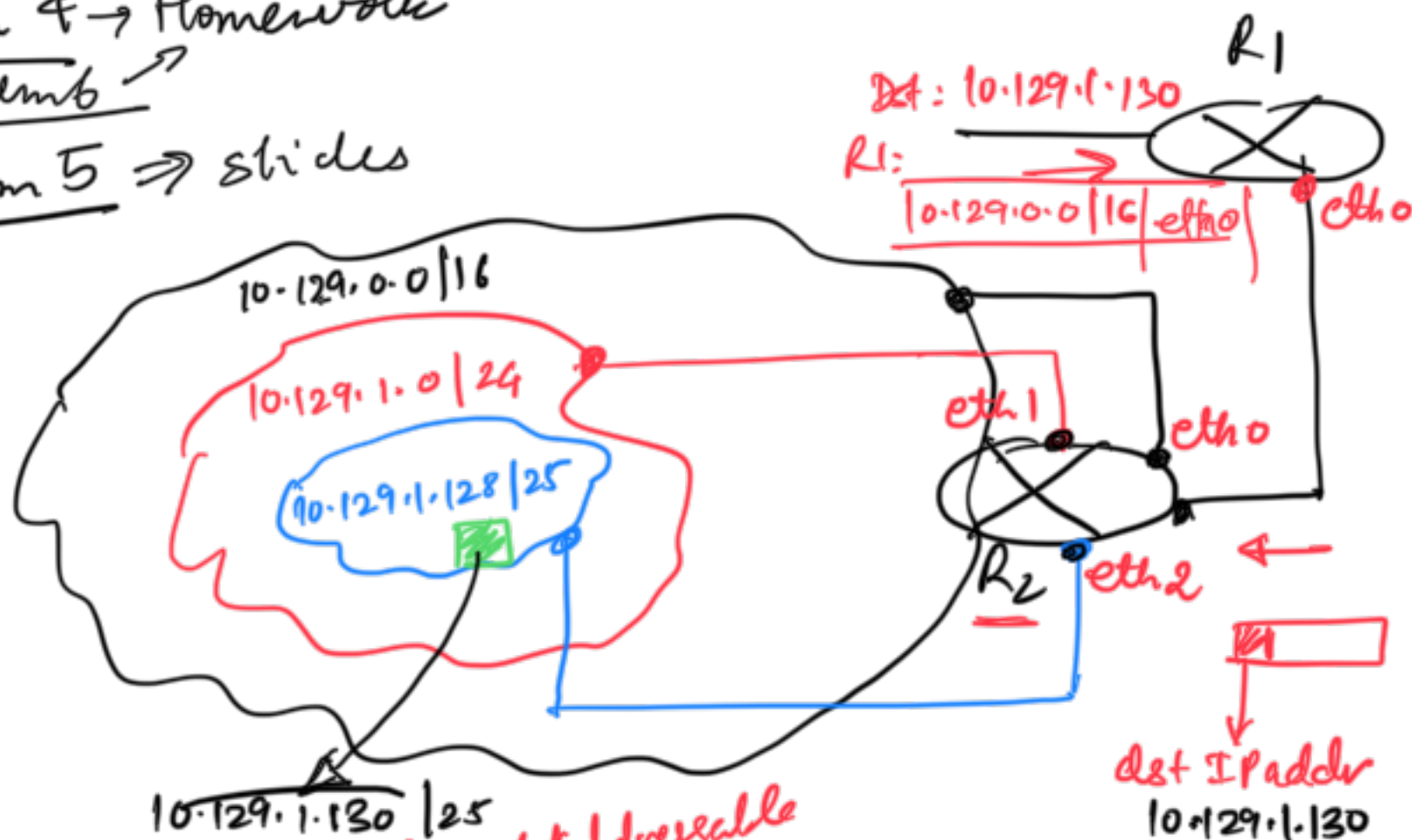
10.129.1.193 to 10.129.1.254

11
11
00000000
11111111

Problem 4 → Homework

Problem 6 →

Problem 5 → slides

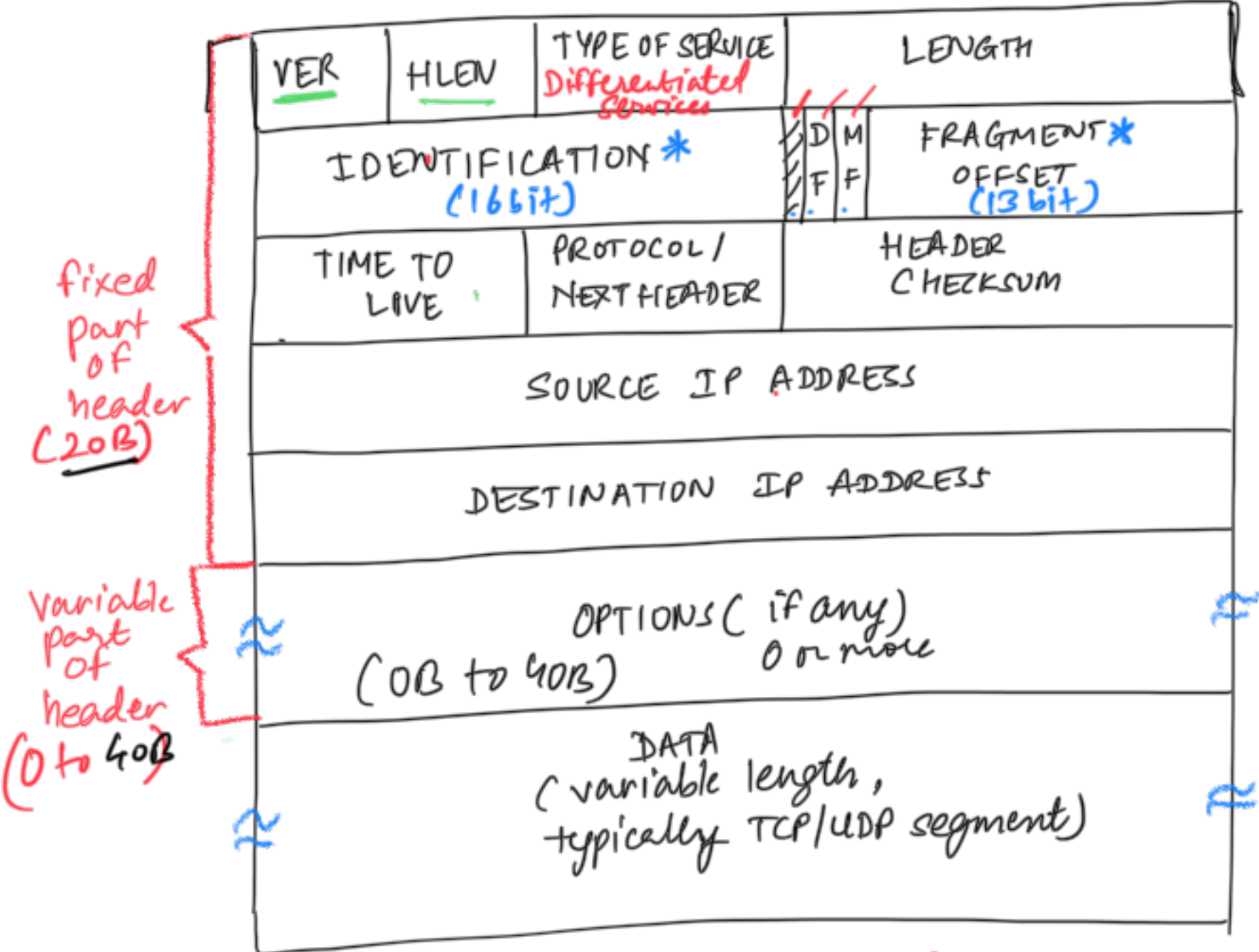
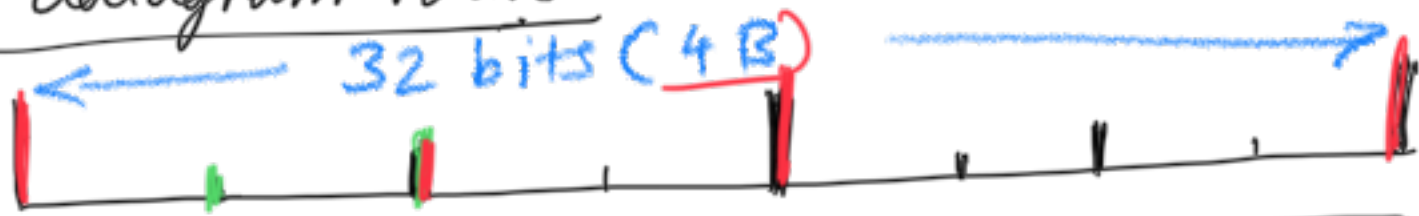


TCAM Ternary Content Addressable Memory
R2's RT Memory

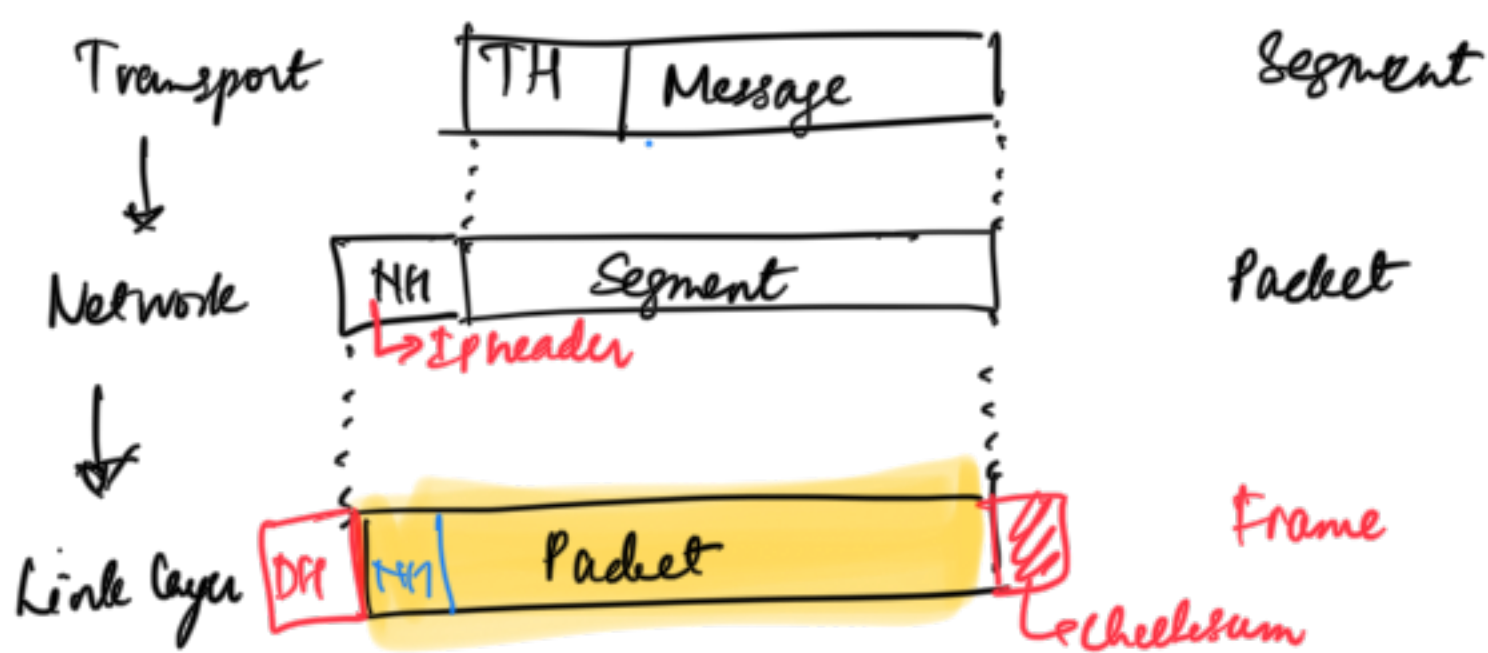
Net addr			
→ 10.129.0.0/16	eth0	✓	^ 255.255.0.0 ⇒ 10.129.0.0
→ 10.129.1.0/24	eth1	✓	^ 255.255.255.0 ⇒ 10.129.1.0
→ 10.129.1.128/25	eth2	✗	^ 255.255.255.128 ↳ 10.129.1.0

Longest Prefix Match (LPM)

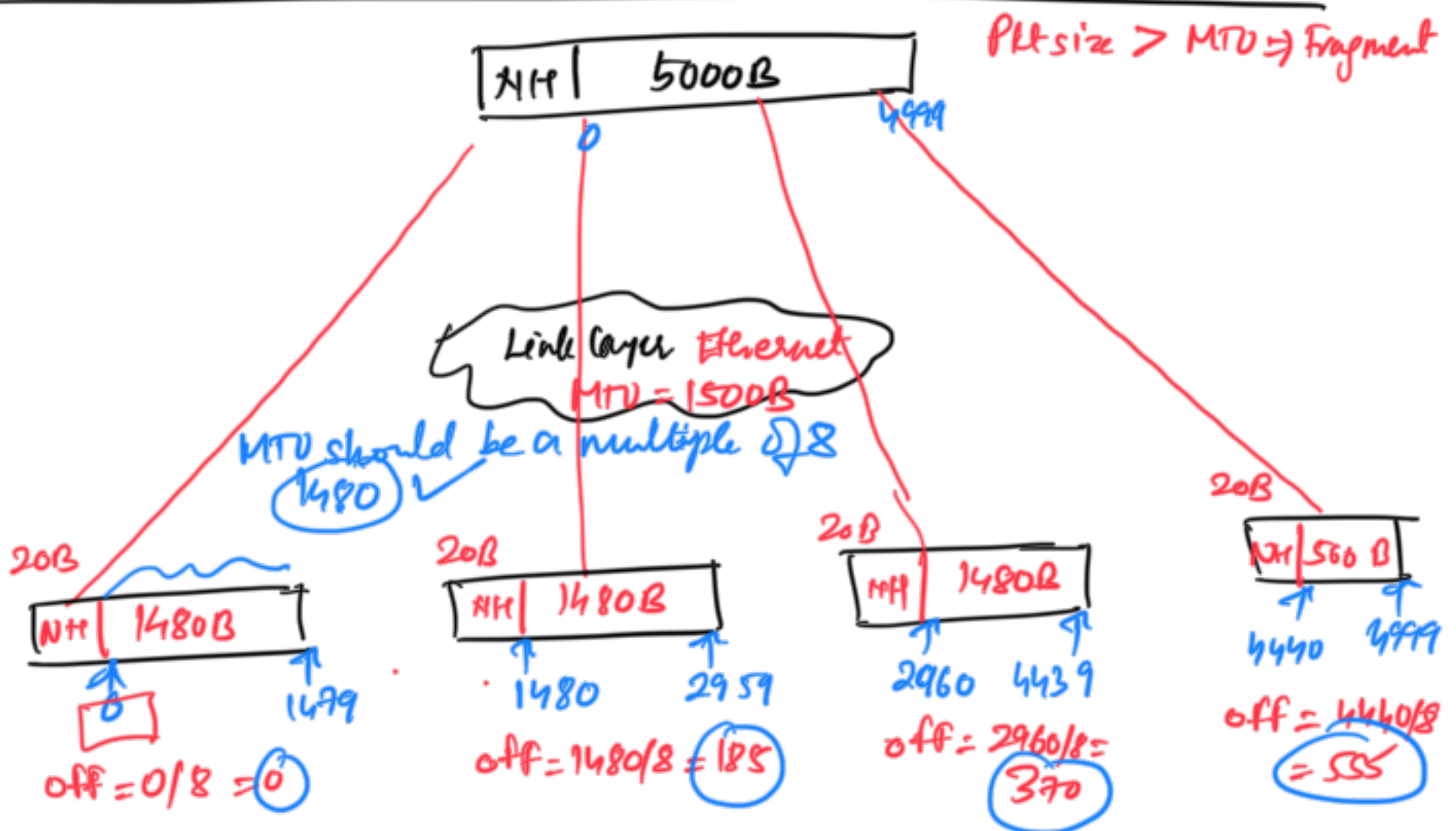
IPv4 datagram format

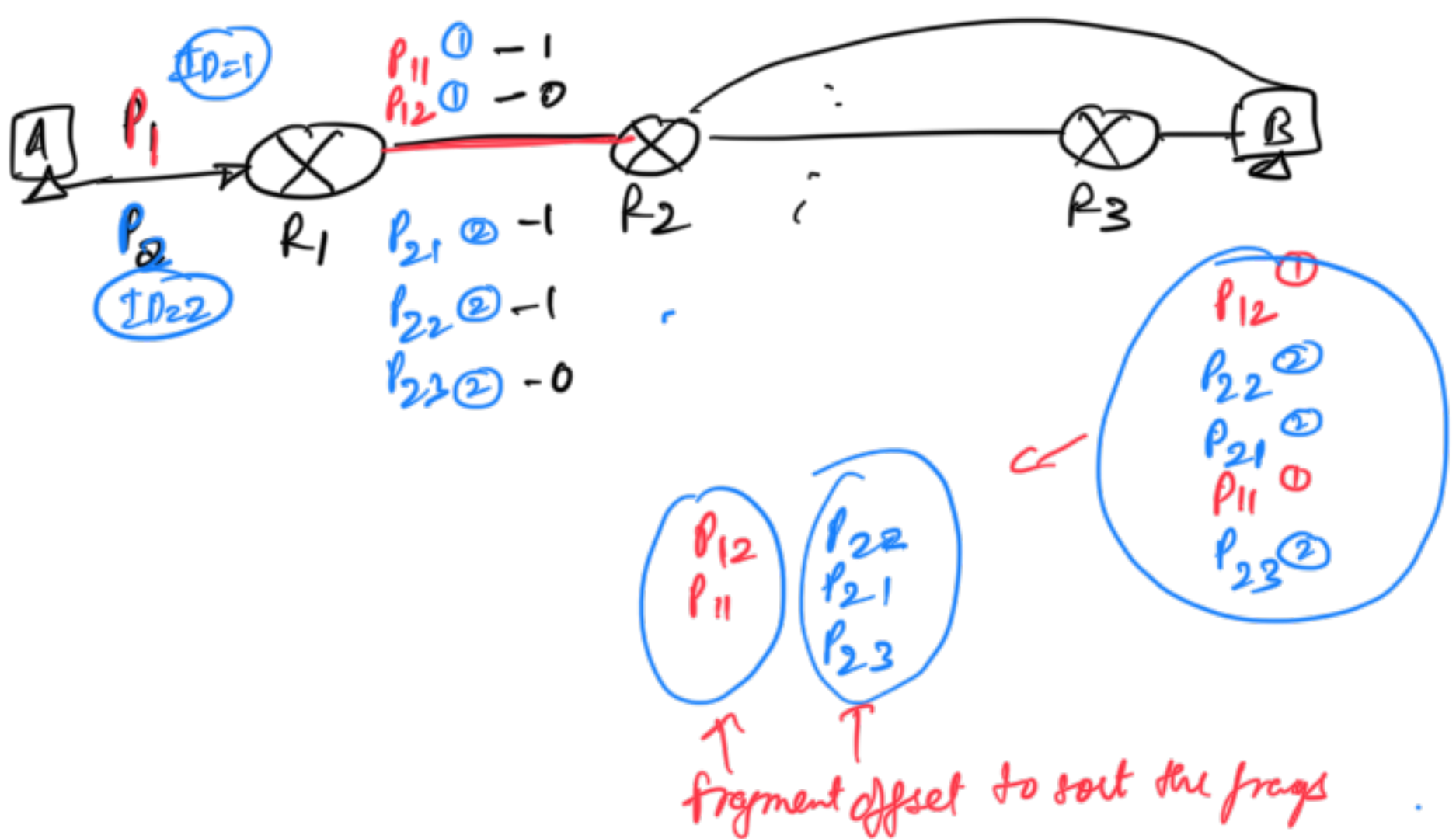


- * VER (4bits) : IP version 0100 $\Rightarrow$ IPV4
- * HLEN (4bits) : Header length in words (1 word = 4B)
 $HLEN_{min} = 0101 = 5 * 4 = 20B$
 $HLEN_{max} = 1111 = 15 * 4 = 60B$
- * TOS (8bits) : 3 bits : Priority (low) to (high) $\rightarrow$ network control pkt
 3 flags : Delay = 1 $\Rightarrow$ low delay
 Throughput = 1 $\Rightarrow$ high



$MTU = 1500B \Rightarrow \text{Ethernet}$
 Max transfer unit
 $MTU = 1480$ excluding IP headers
 $(1500 - 20)$





* **DF** : Don't fragment
(1 bit)

DF = 1 $\Rightarrow$ Don't fragment the pkt

* **MF** : More fragments
(1 bit)

MF = 1 $\Rightarrow$ More frags coming

MF = 0 $\Rightarrow$ last fragment

◦ Length: len (header + data) in bytes
(16b) $HLEN = 5 \text{ words} = 5 \times 4 = 20B$

data len = Length - $HLEN \times 4$
eg. $LEN = 4000, HLEN = 5$
 $DATA LEN = 4000 - 5 \times 4 = 3980B$

◦ Identification: all fragments of same packet have same ID; used to identify fragments of same packet at Rx
(16b)

◦ Flags: DF: Don't Fragment
 $DF = 0$... do not fragment
 $DF = 1$... may fragment

MF: More fragments
 $MF = 1$... more fragments
 $MF = 0$... last fragment

◦ Fragment offset: specifies the offset of fragment in the original packet
(13b) (used for sorting fragments at the Rx)